

In the product $E \times E$, the diagonal and the graph of the endomorphism g will be denoted respectively by Δ and Γ_g . Let

$$\gamma : E \longrightarrow E \times E, \quad \delta : E \longrightarrow E \times E$$

be the corresponding immersions. On the completed local rings at x and at (x, x) , they are given by homomorphisms

$$\gamma^*, \delta^* : \widehat{A} \widehat{\otimes} \widehat{A} \longrightarrow \widehat{A} \quad (A = \widehat{\mathcal{O}}_x),$$

where

$$\delta^*(a \otimes b) = ab, \quad \gamma^*(a \otimes b) = a g^*(b).$$

We have to evaluate the multiplicity $v_x(g)$ of the point $\delta(x)$ in the intersection $\Delta \Gamma_g$ on $E \times E$. The divisor Δ corresponds to the ideal generated by the element $1 \otimes \pi - \pi \otimes 1$ of $\widehat{A} \widehat{\otimes} \widehat{A}$. Hence the multiplicity of $\delta(x)$ in the intersection $\Delta \Gamma_g$ is

$$v_x(g) = v_x(\gamma^*(1 \otimes \pi - \pi \otimes 1)) = v_x(g(\pi) - \pi),$$

as required.

4.2

Let C be a smooth, proper, connected algebraic curve over the algebraically closed field k . If η is its generic point, let $K = k(\eta)$ be its field of rational functions.

Let

$$K' \supset K, \quad [K' : K] = n$$

be a finite Galois extension of degree n with Galois group G . Let C' be the normalization of C in K' , and let

$$p : C' \longrightarrow C$$

be the canonical projection. The Galois group G acts naturally on C' . The stabilizer of $y' \in C'$ in G will be denoted by $G_{y'}$.

If π is a uniformizer of $\mathcal{O}_{C', y'}$ and if $s \in G_{y'}$, $s \neq e$, put

$$v_{y'}(s(\pi) - \pi) = v_{y'}(s) \tag{4.2.1}$$

(the multiplicity of the fixed point y' for the map $s : C' \rightarrow C'$ (Lemma 4.1)).

The Swan representation associated with y' is defined by its character $\sigma_{y'}$, given by

$$\sigma_{y'}(s) = \begin{cases} 1 - v_{y'}(s), & s \neq e, \\ \sum_{s \neq e} (v_{y'}(s) - 1), & s = e. \end{cases} \tag{4.2.2}$$

The results of Artin–Serre–Swan (IX Th. 4) show that, for a prime number ℓ distinct from the characteristic of k , there exists a projective $\mathbb{Z}_\ell[G_{y'}]$ -module of finite type $Sw_{y'}$, determined up to isomorphism, whose character is $\sigma_{y'}$.

It is also useful to consider the $\mathbb{Z}_\ell[G_{y'}]$ -module

$$Sw'_{y'} = Sw_{y'} \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell[G_{y'}]. \tag{4.2.3}$$

The character $\sigma'_{y'}$ of $Sw'_{y'}$ is given by

$$\sigma'_{y'}(s) = \begin{cases} 1 - v_{y'}(s), & s \neq e, \\ 1 + \sum_{s \neq e} v_{y'}(s) = 1 + v_{y'}(\mathfrak{D}_{C'/C}), & s = e, \end{cases} \quad (4.2.4)$$

where $\mathfrak{D}_{C'/C}$ is the different of the covering C'/C ; the last formula is just [3, Chap. VI 2, prop. 4] with $H = \{1\}$.

Finally, the function

$$a_{y'} = \sigma_{y'} + u_{G_{y'}} = \sigma'_{y'} - 1_{G_{y'}}, \quad (4.2.5)$$

where $u_{G_{y'}}$ denotes the character of the augmentation ideal of $\mathbb{Z}_\ell[G_{y'}]$, is also the character of a $\mathbb{Q}_\ell[G_{y'}]$ -module, the Artin representation of $G_{y'}$, which will be denoted by $Art_{y'}$.

Thus

$$a_{y'}(s) = \begin{cases} -v_{y'}(s), & s \neq e, \\ \sum_{s \neq e} v_{y'}(s) = v_{y'}(\mathfrak{D}_{C'/C}), & s = e. \end{cases} \quad (4.2.6)$$

Now consider the inclusion $G_{y'} \subset G$. We put

$$Sw_y = Sw_{y'} \otimes_{\mathbb{Z}_\ell[G_{y'}]} \mathbb{Z}_\ell[G], \quad Sw'_y = Sw'_{y'} \otimes_{\mathbb{Z}_\ell[G_{y'}]} \mathbb{Z}_\ell[G], \quad (4.2.7)$$

$$Art_y = Art_{y'} \otimes_{\mathbb{Q}_\ell[G_{y'}]} \mathbb{Q}_\ell[G]. \quad (4.2.8)$$

The characters of the preceding representations are denoted by σ_y , σ'_y , and a_y .

4.3

The modules so obtained depend, up to isomorphism, only on the point $y \in C$, and not on the choice of the point y' of C' above y . Let y'' be another such point. There exists $g \in G$ such that $y'' = gy'$. The situation (C', y'', G) is deduced from the situation (C', y', G) by the isomorphism g on C' and by $\text{int}(g) : h \mapsto ghg^{-1}$ on G . It follows by transport of structure that the modules Sw_y , etc., associated with y'' are deduced from those associated with y' by extension of the operators by means of $\text{int}(g) : G \rightarrow G$. This extension is known to transform a module into an isomorphic module.

Proposition 4.4. With the notation of 3.7, with $(\Lambda = \mathbb{Z}_\ell)$ and with $M^\bullet$ an object of $D^+(A[G])$, one has isomorphisms

$$\begin{aligned} (*) \quad & Sw_y \simeq \check{S}w_y, \\ (**) \quad & \text{Hom}_{\mathbb{Z}_\ell[G]}^\bullet(Sw_y, M^\bullet) \simeq Sw_y \otimes_{\mathbb{Z}_\ell[G]} M^\bullet. \end{aligned}$$

Proof. For two projective $\mathbb{Z}_\ell[G]$ -modules of finite type to be isomorphic, it is enough that they have the same character (3.2). If $\check{\sigma}_y$ denotes the character of $\check{S}w_y$, then by (3.7.2)

$$\check{\sigma}_y(s) = \sigma_y(s^{-1}).$$

On the other hand

$$v_{y'}(s^{-1}(\pi) - \pi) = v_{y'}(s^{-1}(s(\pi) - \pi)) = v_{y'}(\pi - s(\pi)) = v_{y'}(s(\pi) - \pi),$$

so $Sw_y \simeq \check{S}w_y$, whence formula (*) by definition (4.2.7). Formula (**) follows from 3.8.

4.5

We shall need below an explicit formula for the character of the Swan representation. In fact one has the formulas

$$\sigma_y(s) = \begin{cases} \sum_{\substack{p(z')=y \\ s(z')=z'}} (1 - v_{z'}(s)), & s \neq e, \\ \sum_{p(z')=y} (1 + v_{z'}(\mathfrak{D}_{C'/C})) - n, & s = e, \end{cases} \quad (4.5.1)$$

$$\sigma'_y(s) = \begin{cases} \sum_{\substack{p(z')=y \\ s(z')=z'}} (1 - v_{z'}(s)), & s \neq e, \\ \sum_{p(z')=y} (1 + v_{z'}(\mathfrak{D}_{C'/C})), & s = e, \end{cases} \quad (4.5.2)$$

$$a_y(s) = \begin{cases} - \sum_{\substack{p(z')=y \\ s(z')=z'}} v_{z'}(s), & s \neq e, \\ \sum_{p(z')=y} v_{z'}(\mathfrak{D}_{C'/C}), & s = e. \end{cases} \quad (4.5.3)$$

To deduce these formulas one uses (3.3.1) and takes into account the following fact: if one fixes a point $y'_0 \in C'$, then for every $y' \in C'$ such that $p(y') = p(y'_0)$ one can find an element $g' \in G$ such that $g'(y') = y'_0$. The elements g' form a system of representatives for the right cosets of $G_{y'_0}$ in G , and moreover

$$G_{y'_0} = g' G_{y'} g'^{-1}.$$

This system of representatives can be used in the second member of formula (3.3.1), and one obtains the formulas displayed above.

4.6

The invariants Sw_y , Sw'_y , Art_y can be defined under more general conditions. Let C' be a smooth algebraic curve over k , not necessarily connected. Suppose that the finite group G acts on C' and that there exists an open subset U of $C = C'/G$, dense in C , such that $C'|U$ is a principal covering of U with group G . For a point $y \in C$, one can define the modules Sw_y , Sw'_y , Art_y by proceeding as above. One can reduce to the case where C' is connected in the following way. Let $y' \in C'$ be such that $p(y') = y$, let C'_1 be the irreducible component of C' containing y' , and let C_1 be the irreducible component of C containing y . If $G_{C'_1}$ is the stabilizer of the component C'_1 , and if η'_1 (resp. η_1) is the generic point of C'_1 (resp. C_1), then the extension

$$K'_1 = k(\eta'_1) \supset k(\eta_1) = K_1$$

is Galois with Galois group $G_{C'_1}$ and C'_1 is the normalization of C_1 in K'_1 (SGA 1 I 10). One can therefore apply the preceding considerations to the curves C'_1 , C_1 and the group $G_{C'_1}$. In this way one obtains projective $\mathbb{Z}_\ell[G_{C'_1}]$ -modules of finite type

$$Sw_{G_{C'_1}, y}, \quad Sw'_{G_{C'_1}, y}, \quad Art_{G_{C'_1}, y},$$

and Sw_y , Sw'_y , Art_y are the induced modules associated with the inclusion $G_{C'_1} \subset G$.

Remark 4.6. The modules Sw_y , Sw'_y , Art_y are purely local invariants: they are known from the group $G_{y'}$ acting on the completion of the local ring $\mathcal{O}_{C',y'}$. Moreover $Sw_y = 0$ if and only if $p : C' \rightarrow C$ is tamely ramified at y' , or equivalently, if $G_{y'}$ has order prime to the characteristic of k [3 VI].

5. The Weil formula (compare [3 VI §4])

5.0

All schemes considered in this number will be defined over an algebraically closed base field k .

We shall use the following notation:

$$\Lambda_\nu = \mathbb{Z}/\ell^\nu \mathbb{Z},$$

where ℓ is a fixed prime number distinct from the characteristic of k .

Let $\mathbb{Z}_\ell \rightarrow \Lambda_\nu$ be the canonical homomorphism. It induces a homomorphism

$$\lambda_\nu : \mathbb{Z}_\ell[G] \longrightarrow \Lambda_\nu[G]$$

and consequently [VIII 9] a reduction homomorphism

$$\lambda_\nu^* : K^*(\mathbb{Z}_\ell[G]) \longrightarrow K^*(\Lambda_\nu[G]).$$

The images by λ_ν^* of the elements associated with the modules Sw_y , Sw'_y will be denoted by

$$Sw_y^{\Lambda_\nu}, \quad Sw_y'^{\Lambda_\nu}.$$

The reduction homomorphism λ_ν^* is in fact an isomorphism [IX 1.5]. The inverse of λ_ν^* will be denoted by μ_ν .

The constant sheaf on the scheme X associated with the ring Λ_ν will be denoted by $\Lambda_{\nu,X}$.

Let V be an open subset of X , and let $j : V \rightarrow X$ be the canonical immersion. We put

$$j_!(\Lambda_{\nu,V}) = \Lambda_{\nu,V,X}.$$

Proposition 5.1. Let C' be a smooth proper algebraic curve over the algebraically closed field k , let G be a finite group acting on C' , let $C = C'/G$, let $p : C' \rightarrow C$ be the canonical projection, let U be a dense open subset of C such that $U' = p^{-1}(U)$ is a principal covering of U with group G , let $Y = C - U$, let $Y' = p^{-1}(Y) = C' - U'$, and let $\text{EP}(C)$ be the Euler–Poincaré characteristic of C (equal to $2 - 2g$ if C is connected of genus g). Under these conditions one has the Weil formula:

$$\text{cl}_{K^*(\Lambda_\nu[G])}(\text{R}\Gamma_{C'}(\Lambda_{\nu,U',C'})) = \text{EP}(C) \text{cl}_{K^*(\Lambda_\nu[G])}(\Lambda_\nu[G]) - \sum_{y \in Y} Sw_y'^{\Lambda_\nu}, \quad (5.2)$$

where the first member of (5.2) is defined using 2.1.

Proof. Denote by S_1 (resp. S_2) the first (resp. second) member of the formula to be proved. Since the reduction homomorphism λ_ν^* is in fact an isomorphism, it is enough to prove that

$$\mu_\nu(S_1) = \mu_\nu(S_2).$$

On the other hand, by (3.3.1), the trace homomorphism

$$K^*(\mathbb{Z}_\ell[G]) \xrightarrow{tr} \mathcal{F}(G, \mathbb{Z}_\ell)$$

is a monomorphism. Consequently, it is enough to prove that

$$\mathrm{tr}(\mu_\nu(S_1)) = \mathrm{tr}(\mu_\nu(S_2)).$$

The character $\sigma_2 = \mathrm{tr}(\mu_\nu(S_2))$ is plainly given by

$$\sigma_2 = \mathrm{EP}(C)r_G - \sum_{y \in Y} \sigma'_y, \quad (5.2.1)$$

where σ'_y is the character of Sw'_y ((4.2.7) and (4.2.3)), and r_G is the regular representation.

Now consider the reduction homomorphisms for traces:

$$\mathcal{F}(G, \mathbb{Z}_\ell) \xrightarrow{\lambda_n^*} \mathcal{F}(G, \mathbb{Z}/\ell^n \mathbb{Z}).$$

One plainly has

$$\mathcal{F}(G, \mathbb{Z}_\ell) = \varprojlim_n \mathcal{F}(G, \mathbb{Z}/\ell^n \mathbb{Z}).$$

It is therefore enough to prove, for every n , that

$$\lambda_n^*(\mathrm{tr}(\mu_\nu(S_1))) = \lambda_n^*(\sigma_2).$$

The first member is equal to $\mathrm{tr}(\lambda_n^*(\mu_\nu(S_1)))$, while

$$\lambda_n^*(\mu_\nu(S_1)) = \mathrm{cl}_{K^*(\Lambda_n[G])}(\mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'})),$$

by the Künneth formula (SGA 4 XVIII)

$$\mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'}) \otimes_{\Lambda_n}^L \Lambda_\nu \simeq \mathrm{R}\Gamma_{C'}(\Lambda_{\nu,U',C'}) \quad \text{for } n \geq \nu.$$

Thus we are reduced to proving the equality

$$\mathrm{tr}(\mathrm{cl}_{K^*(\Lambda_n[G])}(\mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'}))) = \lambda_n^*(\sigma_2).$$

Let τ denote the first member of this equality, and let $g \in G$. The element g plainly induces an endomorphism of the distinguished triangle

$$\begin{array}{ccc} & \mathrm{R}\Gamma_{Y'}(\Lambda_{n,Y'}) & \\ & \searrow & \\ \mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'}) & \xrightarrow{\quad\quad\quad} & \mathrm{R}\Gamma_{C'}(\Lambda_{n,C'}) \end{array}$$

Therefore

$$\tau(g) = \mathrm{Tr}^* g_{\mathrm{R}\Gamma_{C'}(\Lambda_{n,U',C'})} = \mathrm{Tr}^* g_{\mathrm{R}\Gamma_{C'}(\Lambda_{n,C'})} - \mathrm{Tr}^* g_{\mathrm{R}\Gamma_{Y'}(\Lambda_{n,Y'})}.$$

For brevity, Tr^* has been written instead of $\mathrm{Tr}_{\Lambda_n}^*$.

Applying a trivial Lefschetz formula to the discrete space Y' on which g acts gives

$$\mathrm{Tr}^* g_{\mathrm{R}\Gamma_{Y'}(\Lambda_{n,Y'})} = \sum_{\substack{y' \in Y' \\ gy' = y'}} 1.$$

On the other hand, the Lefschetz formula (III 4.7) for g acting on C' gives, if $g \neq e$,

$$\mathrm{Tr}^* g_{\mathrm{R}\Gamma_{C'}(\Lambda_{n,C'})} = \sum_{y' \in Y'} v_{y'}(g),$$

the sum of the multiplicities of the fixed points of the map $g_{C'} : C' \rightarrow C'$. Also

$$\mathrm{Tr}^* g_{\mathrm{R}\Gamma_{C'}(\Lambda_{n,C'})} = \mathrm{EP}(C') \cdot 1_{\Lambda} \quad \text{for } g = e,$$

by the explicit computation of $H^*(C', \Lambda_{n,C'})$, via (SGA 4 IX 4) and Künneth (SGA 4 XVII). We therefore obtain the following formula for τ :

$$\tau(g) = \begin{cases} \sum_{\substack{y' \in Y' \\ g(y') = y'}} (v_{y'}(g) - 1), & g \neq e, \\ \mathrm{EP}(C') - \mathrm{card}(Y'), & g = e. \end{cases} \quad (5.2.2)$$

Comparison with (5.2.1) gives easily $\tau = \lambda_n^*(\sigma_2)$. Indeed, if $g \neq e$, the equality between $\tau(g)$ and $\lambda_n^*(\sigma_2)$ is immediate from (4.5.2). If $g = e$, the formula to be checked is

$$\mathrm{EP}(C') - \mathrm{card}(Y') = N \mathrm{EP}(C) - \sum_{y \in Y} \sigma'_y(e), \quad N = \mathrm{card}(G),$$

that is, by (4.5.2),

$$\mathrm{EP}(C') - N \mathrm{EP}(C) = -\deg(\mathfrak{D}_{C'/C}),$$

which is nothing other than the classical Hurwitz formula.

6. Definition of the local terms $\varepsilon_x^\Delta(F)$

6.1

Let C be a connected smooth algebraic curve over the algebraically closed field k , let A be a Λ -algebra ($\Lambda = \mathbb{Z}/\ell^\nu \mathbb{Z}$, with ℓ distinct from the characteristic of k), and let $\Delta \subset D^+(A)$ be a triangulated subcategory of $D^+(A)$ stable under direct factors. Let F be a complex of sheaves of A -modules on C satisfying the following conditions:

- i) For every geometric point $\bar{x} \in C(k)$, one has $F_{\bar{x}} \in \mathrm{ob}(\Delta)$.
- ii) There exists a non-empty open subset of C on which F is locally constant; equivalently, there exist a non-empty open subset U of C and a connected Galois covering $C' \xrightarrow{p} C$ of C , with group G , such that, if $F' = p^*(F)$ is the inverse image of F by p , then $F'|_{U'}$ is constant ($U' = p^{-1}(U)$).

The sheaf $F'|_{U'}$ is therefore defined by a complex M of A -modules on which G acts. If η (resp. η') denotes the generic point of C (resp. C'), and if $\bar{\eta}$ (resp. $\bar{\eta}'$) is a geometric point above η (resp. η'), then

$$p(\eta') = \eta, \quad F_{\bar{\eta}} = F'_{\bar{\eta}'} = M. \quad (6.1.1)$$

As a complex of A -modules, M is isomorphic to $F_{\bar{\eta}}$ and, by condition i), is an object of Δ . Hence, by (3.6), for every $x \in C(k)$,

$$\mathrm{Hom}_{A[G]}^{\bullet}(Sw_x^A, M)$$

is an object of Δ , and so defines an element of $K(\Delta)$.

In fact the object of Δ defined by $\mathrm{Hom}_{A[G]}^{\bullet}(Sw_x^A, M)$ is independent, up to isomorphism, of the Galois covering $C' \rightarrow C$ which trivializes $F|U$. To show this it is enough to compare the elements associated with two coverings

$$C' \xrightarrow{p} C, \quad C'' \xrightarrow{p'} C' \xrightarrow{p} C.$$

In this case, if G' is the Galois group of the covering $C'' \xrightarrow{p''} C$, then G is a quotient group of G' . Let

$$\varphi : \mathbb{Z}_{\ell}[G'] \longrightarrow \mathbb{Z}_{\ell}[G]$$

be the associated canonical epimorphism. The Swan modules are then related by

$$Sw_{G,x} \simeq Sw_{G',x} \otimes_{\mathbb{Z}_{\ell}[G']} \mathbb{Z}_{\ell}[G],$$

which follows immediately from the fact that the character $a_{G,x}$ is the character induced from G' to G ([3] VI Prop. 3).

On the other hand, there is an isomorphism of complexes of $A[G']$ -modules

$$F_{\bar{\eta}'} \simeq [F_{\bar{\eta}''}]_{\varphi}.$$

It follows that there is a canonical isomorphism of complexes of A -modules between

$$\mathrm{Hom}_{A[G]}^{\bullet}(Sw_{G,x}^A, F_{\bar{\eta}}) \quad \text{and} \quad \mathrm{Hom}_{A[G']}^{\bullet}(Sw_{G',x}^A, F_{\bar{\eta}''})$$

(the adjunction formula).

6.2

We shall use the following notation:

$$\alpha_x^{\Delta}(F) = \mathrm{cl}_{K(\Delta)}(\mathrm{Hom}_{A[G]}^{\bullet}(Sw_x^A, F_{\bar{\eta}})), \quad (6.2.1)$$

$$\varepsilon_x^{\Delta}(F) = \alpha_x^{\Delta}(F) + (\mathrm{cl}_{K(\Delta)}(F_{\bar{\eta}}) - \mathrm{cl}_{K(\Delta)}(F_x)). \quad (6.2.2)$$

Of course, in formula (6.2.1), $F_{\bar{\eta}}$ is endowed with its $A[G]$ -module structure by the isomorphism (6.1.1). Note that if $x \in C(k)$ is such that C' is tamely ramified at x , then by (4.6) $Sw_x = 0$, hence $\alpha_x^{\Delta}(F) = 0$, and in this case, in conformity with Ogg-Shafarevich, one has

$$\varepsilon_x^{\Delta}(F) = \mathrm{cl}_{K(\Delta)}(F_{\bar{\eta}}) - \mathrm{cl}_{K(\Delta)}(F_x). \quad (6.2.3)$$

It is clear that $\varepsilon_x^{\Delta}(F) = 0$ if $x \notin Y = C - U$, and therefore only finitely many x satisfy $\varepsilon_x^{\Delta}(F) \neq 0$.

We now give examples of categories Δ and complexes of sheaves F for which the conditions above are satisfied.

a) Let $\Delta = D^b(A)_{\mathrm{coh}}$ (VIII 5), and let F be a constructible complex of sheaves of A -modules (SGA 4 XVII) with bounded cohomology. In this case $\varepsilon_x^{\Delta}(F) \in K_0(A)$.

b) Take $\Delta = D(A)_{\text{parf}}$ (VIII 6), and let F be a constructible complex of sheaves of A -modules of finite tor-dimension. In this case $\varepsilon_x^\Delta(F) \in K^*(A)$.

c) Let A, B be two Λ -algebras and let $\varphi : A \rightarrow B$ be a homomorphism of Λ -algebras. Take $\Delta = D^b(B \text{ rel } A)_{\text{coh}}$ (VIII 10), and let F be a complex of sheaves of B -modules, constructible as a sheaf of complexes of A -modules and with bounded cohomology. One obtains $\varepsilon_x^\Delta(F) \in K_0(B \text{ rel } A)$. Of course, one can vary this as above to obtain, for example, elements $\varepsilon_x^\Delta(F) \in K^*(B \text{ rel } A)$.

7. Euler–Poincaré formula

Theorem 7.1. Let C be a connected, smooth, proper algebraic curve over the algebraically closed field k , let η be the generic point of C , let $\Delta \subset D^+(A)$ be a triangulated subcategory of $D^+(A)$ stable under direct factors, and let F be a complex of sheaves of A -modules satisfying conditions i), ii) of 7.1. Under these conditions one has $R\Gamma_C(F) \in \text{ob}(\Delta)$, and the following formula (Euler–Poincaré formula) is true:

$$\text{cl}_{K(\Delta)}(R\Gamma_C(F)) = \text{EP}(C) \text{cl}_{K(\Delta)}(F_\eta) - \sum_{x \in C(k)} \varepsilon_x^\Delta(F), \quad (7.2)$$

where the $\varepsilon_x^\Delta(F)$ are defined in 6.2 (formula 6.2.2).

The proof of this theorem will be made in a few steps.

a) Let Ψ be a bounded below complex of sheaves of $\Lambda[G]$ -modules on C . There is an isomorphism

$$R\Gamma^G(R\Gamma_C(\Psi)) \simeq R\Gamma_C(R\Gamma^G(\Psi)), \quad (7.3)$$

where Γ^G is the functor “subsheaf of G -invariant sections.” Indeed, Γ_C sends an injective sheaf of $\Lambda[G]$ -modules to an injective $\Lambda[G]$ -module, and Γ^G sends an injective sheaf of $\Lambda[G]$ -modules to an injective sheaf of Λ -modules (using, for example, the evident adjoints of Γ_C and Γ^G). Moreover

$$\Gamma^G \circ \Gamma_C = \Gamma_C \circ \Gamma^G.$$

Taking derived functors on both sides and applying [5] prop. 3.1) gives (7.3).

b) Let Φ be a bounded below complex of sheaves of Λ -modules on C , zero outside U . There is an isomorphism

$$\Phi \simeq R\Gamma^G(p_*(\Phi')), \quad \Phi' = p^*(\Phi). \quad (7.4)$$

First note that the functors p^* and $\Gamma^G \circ p_*$ establish quasi-inverse equivalences between the category of sheaves of Λ -modules on C' on which G acts and which are zero outside $U' = p^{-1}(U)$, and the category of sheaves of Λ -modules on C which are zero outside U . Hence one has the isomorphism

$$\Phi \simeq R((\Gamma^G \circ p_*) \circ p^*)(\Phi).$$

On the other hand, one can again apply [5] Prop. 3.1), which gives, since $Rp^* = p^*$ and $Rp_* = p_*$:

$$\Phi \simeq R(\Gamma^G \circ p_*)(p^*(\Phi)) \simeq (R\Gamma^G)(p_*p^*(\Phi)),$$

which is the desired isomorphism.

c) Let Φ be a complex of sheaves as in b). One has the isomorphism

$$R\Gamma_C(\Phi) = (R\Gamma^G)(R\Gamma_{C'}(\Phi')), \quad \Phi' = p^*(\Phi). \quad (7.5)$$

Indeed,

$$(R\Gamma^G)(R\Gamma_{C'}(\Phi')) \simeq (R\Gamma^G)(R\Gamma_C(p_*(\Phi'))) \simeq R\Gamma_C(R\Gamma^G(p_*(\Phi')))$$

by (7.3), and (7.5) follows from the isomorphism (7.4).

d) Let $i' : U' \rightarrow C'$, $i : U \rightarrow C$ be the canonical inclusions and put

$$\Phi = F_{U,C} = i_!(F|U).$$

To compute $R\Gamma_C(\Phi)$, one may apply formula (7.5). On the other hand, with the notation of 7.1,

$$\Phi' \simeq \Lambda_{U',C'} \otimes_{\Lambda}^L M,$$

and the Künneth formula (SGA 4 XVII) gives

$$R\Gamma_{C'}(\Phi') \simeq R\Gamma_{C'}(\Lambda_{U',C'} \otimes_{\Lambda}^L M) \simeq R\Gamma_{C'}(\Lambda_{U',C'}) \otimes_{\Lambda}^L M. \quad (7.6)$$

This last isomorphism is in fact an isomorphism between the objects of $D(A[G])$ defined by the two sides. Combining (7.6), (7.5), and formula (3.9.1), which is applicable by 2.1, one obtains

$$R\Gamma_C(F_{U,C}) \simeq R\Gamma^G(R\Gamma_{C'}(\Lambda_{U',C'}) \otimes_{\Lambda}^L M) \simeq R\mathrm{Hom}_{A[G]}(R\Gamma_{C'}(\Lambda_{U',C'})^{\vee}, M). \quad (7.7)$$

Thus, since $M \simeq F_{\bar{\eta}} \in \mathrm{ob}(\Delta)$ (condition (i)) and by (3.6), we find

$$R\Gamma_C(F_{U,C}) \in \mathrm{ob}(\Delta).$$

On the other hand, there is the distinguished triangle (where $Y = C - U$)

$$\begin{array}{ccc} & R\Gamma_Y(F|Y) & \\ & \searrow & \\ R\Gamma_C(F_{U,C}) & \xrightarrow{\quad\quad\quad} & R\Gamma_C(F) \end{array} \quad (7.8)$$

where obviously

$$R\Gamma_Y(F) = \sum_{y \in Y} F_y \in \mathrm{ob}(\Delta),$$

because $F_y \in \mathrm{ob}(\Delta)$ (condition (i)). It follows that $R\Gamma_C(F) \in \mathrm{ob}(\Delta)$, which is the first assertion of the theorem. The distinguished triangle (7.8) furthermore gives the relation

$$\mathrm{cl}_{K(\Delta)}(R\Gamma_C(F)) = \mathrm{cl}_{K(\Delta)}(R\Gamma_C(F_{U,C})) + \sum_{y \in Y} \mathrm{cl}_{K(\Delta)}(F_y),$$

and, using the isomorphism 7.7,

$$\mathrm{cl}_{K(\Delta)}(R\Gamma_C(F)) = \mathrm{cl}_{K(\Delta)}(R\mathrm{Hom}_{A[G]}(R\Gamma_{C'}(\Lambda_{U',C'})^{\vee}, F_{\bar{\eta}})) + \sum_{y \in Y} \mathrm{cl}_{K(\Delta)}(F_y).$$

To obtain the Euler–Poincaré formula it remains only to use the Weil formula (5.2). Indeed, first 4.4, (5.2), and 3.6 (in the particular case $B = A[G]$) give

$$\mathrm{cl}_{K(\Delta)}(R\Gamma_C(F)) = \sum_{y \in Y} \mathrm{cl}_{K(\Delta)}(F_y) + \mathrm{EP}(C) \mathrm{cl}_{K(\Delta)}(F_{\bar{\eta}}) - \sum_{y \in Y} (\alpha_y^{\Delta}(F) + \mathrm{cl}_{K(\Delta)}(F_{\bar{\eta}})).$$

Since $\varepsilon_x^\Delta(F) = 0$ for $x \notin Y$, this is exactly formula (7.2). For a complex of sheaves F satisfying the conditions of 7.1, we introduce the notation

$$\text{EP}_\Delta(C, F) = \text{cl}_{K(\Delta)}(\text{R}\Gamma_C(F)).$$

Corollary 7.9. If F, F' are complexes of sheaves satisfying the conditions of 7.1 and if they are locally isomorphic (for the étale topology) as complexes of A -modules, then

$$\text{EP}_\Delta(C, F) = \text{EP}_\Delta(C, F'). \quad (7.10)$$

In particular, if F is locally constant of value $M^\bullet$, then

$$\text{EP}_\Delta(C, F) = \text{EP}(C) \text{cl}_{K(\Delta)}(M^\bullet). \quad (7.11)$$

Corollary 7.12. Let C be a connected smooth proper algebraic curve over k , let V be a non-empty open subset of C , let Δ be a subcategory of $D^+(A)$ as in 7.1, and let F be a complex of sheaves of A -modules on V such that its restriction to a suitable non-empty open subset of V is trivialized by a Galois covering of V . Suppose moreover that $F_{\bar{x}} \in \text{ob}(\Delta)$ for every $x \in V$. Under these conditions one has

$$\begin{aligned} (*) \quad & \text{R}\Gamma_! V(F), \text{R}\Gamma_V(F) \in \text{ob}(\Delta), \\ (**) \quad & \text{cl}_{K(\Delta)}(\text{R}\Gamma_! V(F)) = \text{cl}_{K(\Delta)}(\text{R}\Gamma_V(F)). \end{aligned}$$

Put $\text{EP}_\Delta(V, F) = \text{cl}_{K(\Delta)}(\text{R}\Gamma_V(F))$ and

$$\text{EP}(V) = \text{EP}_{D(F_\ell)_{\text{coh}}}(V, F_\ell) = \text{EP}(C) - \text{card}(Z),$$

where $Z = C - V$ and where F_ℓ is the sheaf on C defined in dimension zero by the constant sheaf associated with $F_\ell = \mathbb{Z}/\ell\mathbb{Z}$. Then one has the formula

$$\text{EP}_\Delta(V, F) = \text{EP}(V) \text{cl}_{K(\Delta)}(F_{\bar{\eta}}) - \sum_{x \in V(k)} \varepsilon_x^\Delta(F) - \sum_{x \in Z} \alpha_x^\Delta(F), \quad (7.13)$$

where ε_x^Δ and α_x^Δ are defined by (6.2.1), (6.2.2).

We shall use the following lemma.

Lemma 7.14. Let K be the field of fractions of a strictly local discrete valuation ring R , let $W = \text{Spec } K$, let A be a noetherian ring annihilated by an integer n prime to the residual characteristic p of R , and let F be a complex of sheaves of A -modules on W trivialized by a finite extension of K . If Δ is a triangulated subcategory of $D^+(A)$ stable under direct factors and if $F_{\bar{\eta}} \in \text{ob}(\Delta)$ (η the generic point of S), then $\text{R}\Gamma_W(F) \in \text{ob}(\Delta)$ and

$$\text{cl}_{K(\Delta)}(\text{R}\Gamma_W(F)) = 0.$$

Proof of 7.14. Let π be the fundamental group of W . Then F corresponds to a complex of A -modules $M^\bullet$ with π as group of operators. The underlying complex of A -modules of $M^\bullet$ is isomorphic to $F_{\bar{\eta}}$ and consequently belongs to $\text{ob}(\Delta)$. Introduce the quotient group $\bar{\pi}$ of π by its unique Sylow p -subgroup P [4, II 33], that is, the projective limit of discrete quotient groups of π which are of

order prime to p , where p is the characteristic exponent of the residue field of R (if $p = 1$, put $\bar{\pi} = \pi$). Thus there is an exact sequence

$$1 \longrightarrow P \longrightarrow \pi \longrightarrow \bar{\pi} \longrightarrow 0,$$

and one obtains

$$R\Gamma_W(F) \simeq R\Gamma^\pi(M^\bullet) \simeq R\Gamma^{\bar{\pi}}(R\Gamma^P(M^\bullet))$$

(Hochschild–Serre isomorphism). Now P acts on $M^\bullet$ through a finite quotient group, which is therefore a p -group, hence of order prime to n . Thus

$$R\Gamma^P(M^\bullet) \simeq \Gamma^P(M^\bullet),$$

and $\Gamma^P(M^\bullet)$ is a direct factor of $M^\bullet$, hence belongs to $\text{ob}(\Delta)$. On the other hand

$$\bar{\pi} = \prod_{\ell \neq p} \mathbb{Z}_\ell$$

([4], II 33). We are therefore reduced to proving that if $M^\bullet \in \text{ob}(\Delta)$, then $R\Gamma^{\bar{\pi}}(M^\bullet) \in \text{ob}(\Delta)$ and

$$\text{cl}_{K(\Delta)}(R\Gamma^{\bar{\pi}}(M^\bullet)) = 0.$$

For this, let f be a topological generator of $\bar{\pi}$. We may suppose that the components M^i of M are $\bar{\pi}$ -modules such that

$$H^j(\bar{\pi}, M^i) = 0 \quad \text{if } j \geq 1.$$

There is then a distinguished triangle

$$\begin{array}{ccc} & M^\bullet & \\ & \searrow 1-f & \\ R\Gamma^{\bar{\pi}}(M^\bullet) & \xrightarrow{\quad} & M^\bullet. \end{array} \tag{7.15}$$

Indeed, on the one hand it is clear that $R\Gamma^{\bar{\pi}} = \text{Ker}(1 - f)$. On the other hand $1 - f : M^\bullet \rightarrow M^\bullet$ is surjective. Indeed

$$H^1(\bar{\pi}, M) = M^\bullet / (1 - f)M^\bullet$$

(see for this calculation [4] pp. 197, in a substantially analogous case), and therefore $M^\bullet = (1 - f)M^\bullet$. The triangle (7.15) gives at the same time $R\Gamma^{\bar{\pi}}(M^\bullet) \in \text{ob}(\Delta)$ and

$$\text{cl}_{K(\Delta)}(R\Gamma^{\bar{\pi}}(M^\bullet)) = 0,$$

which proves 7.14.

Proof of 7.12. The fact that $R\Gamma_! V(F) \in \text{ob}(\Delta)$ follows immediately from 7.1 applied to the complex $i_!(F)$, where $i : V \rightarrow C$; indeed

$$R\Gamma_! V(F) = R\Gamma_C(i_! F).$$

To prove that $R\Gamma_V(F) \in \text{ob}(\Delta)$, consider the distinguished triangle

$$\begin{array}{ccc} & Q^\bullet & \\ & \searrow & \\ R i_!(F) & \xrightarrow{\quad} & R i_*(F), \end{array}$$

where $Q^\bullet$ is concentrated on $Z = C - V$. Applying $R\Gamma_C$ gives the distinguished triangle

$$\begin{array}{ccc} & R\Gamma_Z(Q^\bullet|Z) & \\ & \searrow & \\ R\Gamma_!V(F) & \xrightarrow{\quad\quad\quad} & R\Gamma_V(F). \end{array} \quad (7.16)$$

Since $R\Gamma_!V(F) \in \text{ob}(\Delta)$, it follows that $R\Gamma_V(F) \in \text{ob}(\Delta)$ if and only if $R\Gamma_Z(Q^\bullet|Z) \in \text{ob}(\Delta)$, that is, if and only if $(Ri_*(F))_{\bar{x}} \in \text{ob}(\Delta)$ for every $x \in Z$. Recall that to calculate $(Ri_*(F))_{\bar{x}}$, one considers the cartesian square

$$\begin{array}{ccc} \bar{V} & \xrightarrow{j} & C_x \\ u \downarrow & & \downarrow \\ V & \longrightarrow & C \end{array}$$

where C_x is the strictly localized scheme corresponding to the geometric point x . One has

$$Ri_*(F)_{\bar{x}} \simeq R\Gamma_{\bar{V}}(u^*(F)),$$

by virtue of 7.14, and therefore $(Ri_*(F))_{\bar{x}} \in \text{ob}(\Delta)$ and

$$\text{cl}_{K(\Delta)}((Ri_*(F))_{\bar{x}}) = 0.$$

It follows that

$$\text{cl}_{K(\Delta)}(R\Gamma_Z(Q^\bullet|Z)) = 0,$$

and the distinguished triangle (7.16) gives

$$\text{cl}_{K(\Delta)}(R\Gamma_!V(F)) = \text{cl}_{K(\Delta)}(R\Gamma_V(F)).$$

Finally, to establish (7.13), it is enough to use the Euler–Poincaré formula (7.2) for the complex of sheaves $i_!(F)$:

$$\text{EP}_\Delta(V, F) = \text{cl}_{K(\Delta)}(R\Gamma_!V(F)) = \text{cl}_{K(\Delta)}(R\Gamma_C(i_!(F))) = \text{EP}(C) \text{cl}_{K(\Delta)}(F_{\bar{\eta}}) - \sum_{x \in C(k)} \varepsilon_x^\Delta(i_!(F)).$$

But

$$\sum_{x \in C(k)} \varepsilon_x^\Delta(i_!(F)) = \sum_{x \in V(k)} \varepsilon_x^\Delta(F) + \sum_{x \in Z} \varepsilon_x^\Delta(F).$$

For $x \in Z$ one has $(i_!(F))_{\bar{x}} = 0$, and consequently, for such x ,

$$\varepsilon_x^\Delta(i_!(F)) = \alpha_x^\Delta(F) + \text{cl}_{K(\Delta)}(F_{\bar{\eta}}),$$

which gives formula (7.13).

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